



Assignment– Implementing Database Operations using Python

Problem Statement

Create Python scripts that connect to a PostgreSQL database and perform basic database operations using the psycopg2 library.

The assignment should cover:

- Connecting to a PostgreSQL database from Python
- Creating a database and a table
- Inserting data into the table
- Fetching and displaying data from the table
- Taking user input and inserting it dynamically using parameterized queries
- Performing operations like SELECT with WHERE conditions, and TRUNCATE

Use the following:

- psycopg2 library for database connection
 - cursor.execute() for running SQL queries
 - conn.commit() and conn.close() for finalizing operations
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Topics Covered

- Connecting Python with PostgreSQL
 - Creating tables
 - Inserting records
 - Fetching records (fetchone)
 - Taking user input and using parameterized queries
 - Creating and dropping databases using psql
 - Creating tables and inserting records using psql
 - SELECT queries with WHERE conditions
 - TRUNCATE TABLE
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Expected Output

- Running the Python scripts should:
 - Create a table successfully in PostgreSQL
 - Insert records into the table
 - Display fetched records in the terminal
 - Show success messages for each operation performed
 - Screenshots of all tasks (code + output) must be included in submission
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Submission Guidelines:

- Create **ONE main folder**.
- Example folder name:
DatabaseOperations-YourName
- Inside this folder include:
 - Python source code file(s) (.py)
 - Screenshots of the code written in VS Code and the corresponding output for each task
- Convert the folder into **ZIP format**.
- Upload the ZIP folder on the submission portal.

If you have any doubt, you can directly contact the Tutedude mentor.